Hennepin County, MN

Roads and rails in proximity to water

This map shows the major interstates, county roads, and railways near bodies of water located within Hennepin County in Minnesota. Transportation and proximity to water have been key aspects of Minnesota's urban geography, so I wanted to demonstrate their interconnectedness particularly in the county where Minneapolis is located. Minneapolis was established on the Mississippi River for its access to both barges and rail as a means for industrial transport, so I wanted to explore the expansiveness of rails and roads as modern forms of transport near the water. All water bodies exemplified in the map are Area Hydrography.

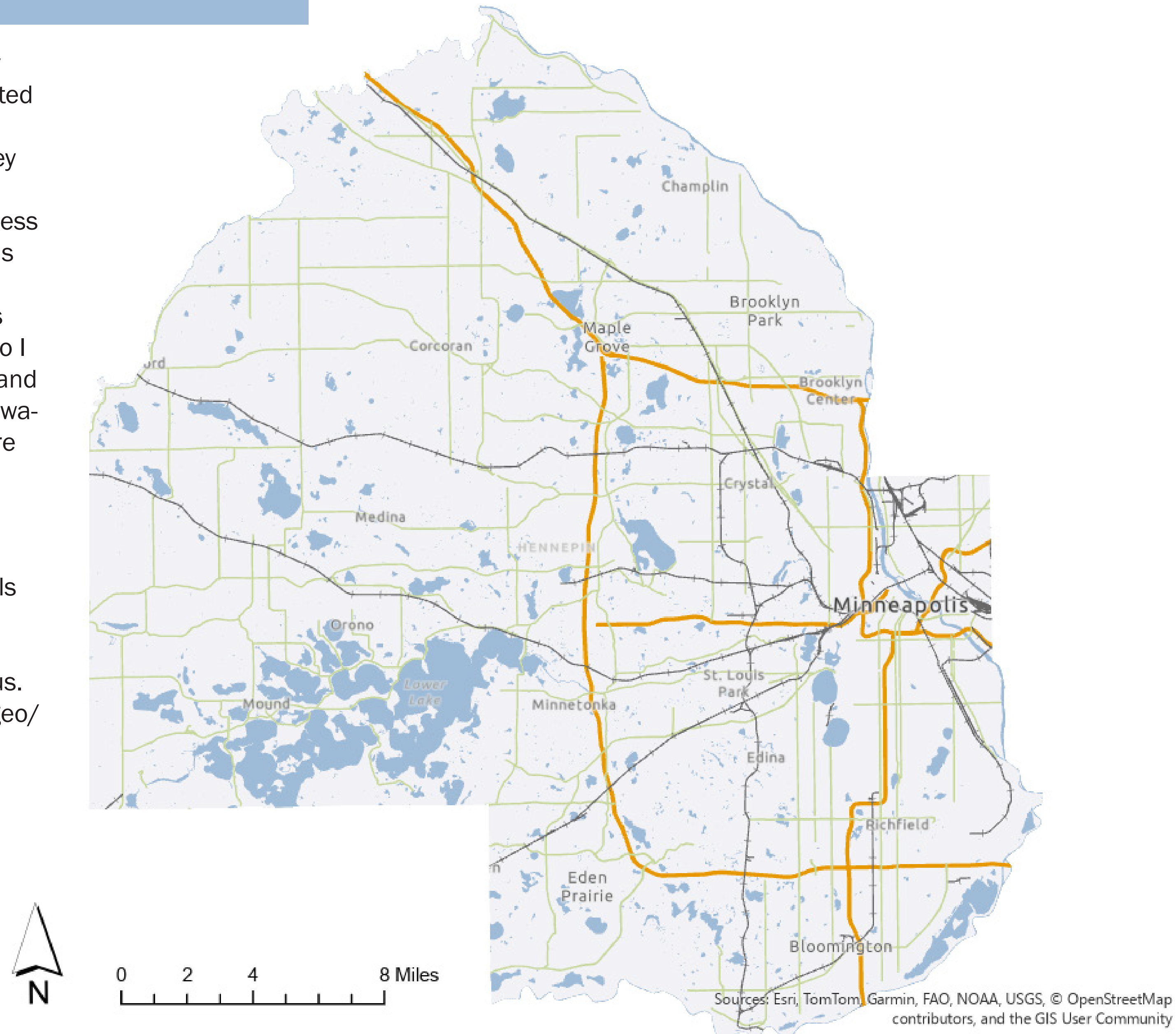
Data Sources

All data, including water areas, roads, and rails from 2022 U.S. Census TIGER files.

"TIGER/Line® Shapefiles." 2024. Www.census.gov. 2024. https://www.census.gov/cgi-bin/geo/shapefiles/index.php.

Legend

- railways
- county roads
- interstate
- water bodies
- hennepin county



1. I am somewhat satisfied with my submission of this map. I am content with how I was able to navigate in ArcGIS Pro to adjust the symbology and customize my map layout to make the overall presentation. However, one of the major issues I ran into was that the data I wanted to display ended up either having issues or simply not existing within the shapefile I downloaded. For example, at first I had wanted to display golf courses in proximity to area hydrography across Hennepin County, but when I opened the attribute table, it showed golf courses listed as the wrong MFCC code. When I looked to display county parks instead, there were only two that showed up, which I knew wasn't true. So, instead, I chose to focus on railway data, as I could visibly tell that it was more accurate than the golf course data I had pulled. If I had more time and data analysis skills, I would have done a deeper dive into why I was struggling to find the original data I was looking to display, and whether this was an issue due to the data attribute filtering I was doing or if it was more of a matter of the data not existing through the TIGER files.

2. In order to complete this assignment, I utilized a few different resources. I first consulted the course skills tutorials, and when I reached a point where the tutorials could not answer my question, I went to either Sarah, the TA, or to my fellow classmates to see how they had navigated around similar issues. I did not have any issues with accessing and filtering the data, but as I mentioned above, struggled a bit when the data wasn't displaying how I had thought it would. In particular, playing around in ArcGIS and consulting with my classmates was immensely helpful because many of us seemed to be running into similar problems, so if one person was able to resolve an issue with their data they were able to assist others to do the same.

3. I used ArcGIS Pro to generate and customize the map layout, and then I used InDesign to add additional text and the title. All data was accessed via the U.S. Census TIGER shapefiles.

4. Yes, I give you my permission to post my assignment to the course website as an example.